Biology Class 11 National Book Foundation FBISE Quality Notes
Studybeesofficial

Chapter 7: Protists and Fungi

7.1 Protists - evolutionary relationship (SLO 1)

- Protists: diverse, mostly aquatic eukaryotes, single celled or simple multicellular; up to **200,000 species**. Greek "protist" = the very first. They are **defined by exclusion**: any eukaryote that is not a plant, animal or fungus.
- **Eukaryotic cells** (nucleus, mitochondria, plastids) separate them from bacteria.

Polyphyletic: A group whose members do not share a common evolutionary ancestor; they evolved independently from different ancestral sources. Protists have many origins (maybe as many as twenty groups).

Feature	Diversity in protists
Size	Microscopic protozoa to giant kelp up to 60 m; unicellular, colonial or coenocytic; no specialized tissues.
Nutrition	Autotrophs (algae), absorptive (water moulds), ingestive (protozoa, slime moulds).
Habitat	Mostly water (plankton); damp soil, leaf litter; free living, mutualistic or parasitic; extremophiles in snow, hot springs, acidic ponds.
Reproduction	Asexual, or sexual with meiosis and syngamy; no multicellular sex organs, no embryos.
Locomotion	Amoeboid motion (pseudopodia), cilia, flagella, or a mix.

7.2 Major groups of protists (SLO 2)

- Four major groups: **Protozoa, Algae, Myxomycota, Oomycota**.

7.2.1 Protozoa - animal-like protists

- Greek proto = first, zoa = animals. Polyphyletic. Aquatic (Amoeba, Paramecium, Radiolaria), parasitic (Plasmodium, Entamoeba histolytica) or in soil / leaf litter (Spiromonas, Vorticella).

Follow Studybeesofficialpak on YouTube Studybeesofficial

- ***Unicellular**; one cell does all life functions. **Vacuoles:** food (digestion), contractile (freshwater: pump out excess water, osmoregulation), storage. Some have shells (foraminifera). Reproduce asexually and sexually; regeneration common; form **resistant cysts** in unfavourable conditions.*
- *Locomotion: pseudopodia (Amoeba), cilia (Paramecium), flagella (Trypanosoma); parasites like Plasmodium have no specific organ.*

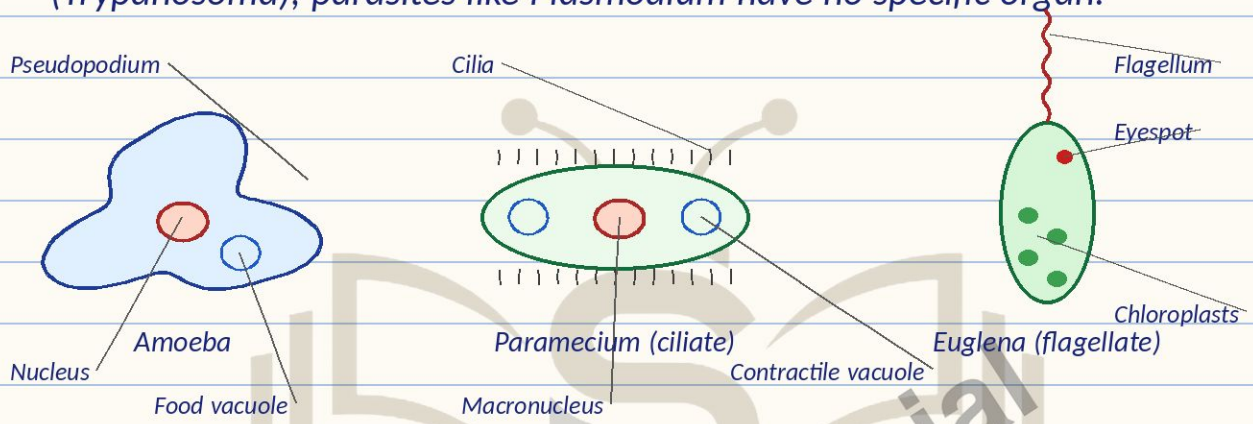


Fig: Amoeba, Paramecium and Euglena

Protozoan group	**Features and examples**
Amoebas	*Pseudopodia for movement and feeding; Amoeba proteus, Entamoeba histolytica.*
Zooflagellates	*Flagella; Giardia intestinalis (small intestine, giardiasis).*
Choanoflagellates	*Sessile on a stalk; flagellum with a collar like sponge collar cells.*
Ciliates	*Covered with cilia (Latin = eyelash); Paramecium, Stentor, Vorticella.*
Foraminifera / Actinopods	*Marine; protective shells (tests): calcium in forams, silica in actinopods.*
Apicomplexans	*Parasitic with complex life cycles and host-invading structures; Plasmodium (malaria).*

- ***Leishmania** (sandfly bite) causes leishmaniasis in Pakistan: skin sores, fever, fatigue.*

7.2.2 Algae - plant-like protists

- *Photosynthetic, in oceans, ponds, hot springs, polar ice, moist soil; responsible for **50-60% of photosynthesis** on earth. Unicellular, filamentous (**coenocytes**:*

Follow Studybeesofficialpak on YouTube Studybeesofficial

*many nuclei, no cross walls) or multicellular (branched or leaf-like **thallus**). No roots, stems, leaves, vascular tissue.*

- *Pigments: chlorophyll a, carotenoids, xanthophyll, phycoerythrin. Almost all (except red algae) have flagellated motile stages. **Unlike plants:** unicellular sex organs, zygote not protected by parent, no embryo.*

Phylum	**Features and examples**
Euglenoids	*Freshwater; one third have chloroplasts, others ingest / absorb; Euglena.*
Dinoflagellates	*Cellulose plates with silica; two flagella; yellow, green, brown; glow (bioluminescent).*
Diatoms	*Most numerous unicellular algae of oceans; box-like silica wall of two halves (lid); frustules.*
Brown algae	*Cool marine waters; up to 75 m; giant kelps.*
Red algae	*Warm seawater, up to 1 m; attached by holdfast; no flagellated cells.*
Green algae	*Cellulose wall and chlorophyll a, b like plants (closest to first plants); Chlamydomonas, Spirogyra, Volvox, Chlorella, Ulva.*

7.2.4 Fungus-like protists

***Myxomycota (plasmodial slime moulds):** Exist as plasmodium = diploid multinucleate cytoplasmic mass wrapped in a slime sheath. In dry (unfavourable) conditions it forms sporangia in which meiosis makes haploid spores.*

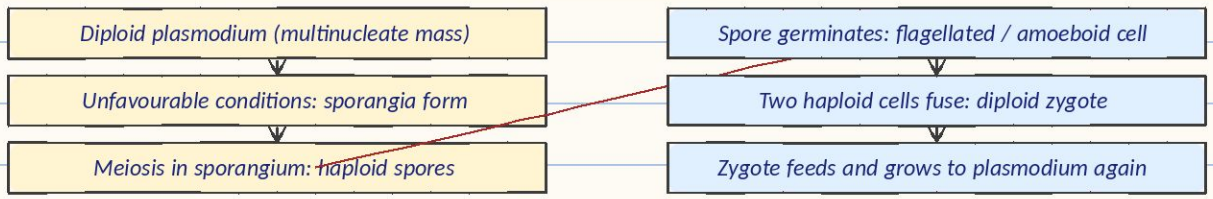


Fig: Life cycle of plasmodial slime mould (Physarum)

Follow Studybeesofficialpak on YouTube Studybeesofficial

- *Spores release a haploid flagellated or amoeboid cell; two fuse to form a diploid zygote that feeds and grows into plasmodium again (Physarum).*

Oomycota (water moulds, white rusts, downy mildews): *Aquatic or moist habitats; parasites or saprotrophs. Cell wall of* ***cellulose*** *(fungi: chitin);* ***aseptate coenocytic hyphae****.*

- ***Asexual:*** *sporangia produce* ***zoospores*** *with a single flagellum; they germinate to hyphae.* ***Sexual:*** *female* ***oogonium*** *(1-8 eggs) and male* ***antheridium*** *(many male nuclei); fusion gives thick walled* ***oospore*** *(survival; gives the phylum its name). Examples:* ***Phytophthora infestans****, Saprolegnia.*

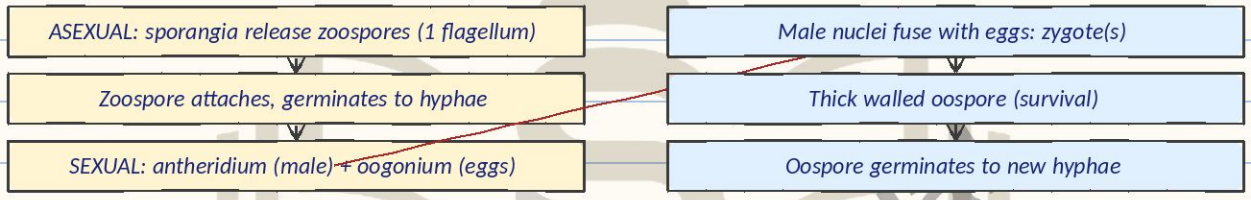


Fig: Life cycle of Oomycota (Phytophthora infestans)

7.2.6 Importance of protists to humans (SLO 3)

Protist	***Importance***
Dinoflagellates	*Red tides (toxic blooms); Alexandrium neurotoxin causes paralytic shellfish poisoning (blocks nerve signals, paralysis, respiratory failure); also primary producers.*
Diatoms	*Major oxygen producers; base of food webs; silica frustules used in cosmetics, pharmaceuticals, nanotechnology.*
Brown algae	*Edible kelp; alginates as thickeners / stabilizers; kelp forests prevent coast erosion; marine biodiversity.*
Red algae	*Food (sushi), cosmetics; agar for culture media and dental impressions; biotechnology research.*

Follow Studybeesofficialpak on YouTube Studybeesofficial

Protist	Importance
Green algae	Producers; Chlorella: photosynthesis research, high-protein health food (Taiwan, Japan); biofuel source.
Plasmodium	Malaria.
Entamoeba histolytica	Amoebic dysentery (contaminated food / water): severe diarrhoea, abdominal pain, fever.
Acanthamoeba	Eye infection (keratitis) from contact lens hygiene, skin and CNS infections.
Phytophthora infestans	Irish potato famine; late blight of potato: brown dead areas on leaflets, haustoria, tubers infected; control: destroy infected plants before lifting tubers.

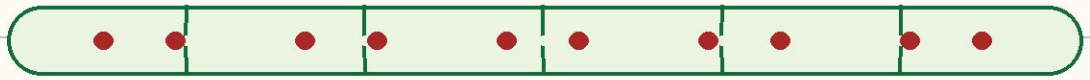
7.3 General characteristics of fungi (SLO 4)

- Eukaryotic, no roots, stems, leaves; study = **mycology**. **Heterotrophs:** no chlorophyll; **extracellular digestion** (enzymes secreted, products absorbed through cell wall). Modes: saprophytic, mutualistic (mycorrhiza), parasitic, predatory.
- **Size:** yeasts a few to 10 micrometre; mushroom fruiting bodies mm to cm; **Armillaria ostoyae** (honey mushroom, Oregon) covers over 2385 acres, the largest living organism.
- **Cell wall: chitin** (nitrogenous polysaccharide). Reserve food: **glycogen** (like animals, not starch).
- **Thallus: mycelium** = network of **hyphae** (increase absorptive surface). **Aseptate** hyphae = coenocytic multinucleate (Rhizopus); **septate** (Penicillium).

Follow Studybeesofficialpak on YouTube Studybeesofficial



Aseptate (coenocytic) hypha: no cross walls, many nuclei - Rhizopus



Septate hypha: cross walls (septa), perforated - Penicillium

Hyphae together form the mycelium (thallus); cell wall = chitin

Fig: Types of fungal hyphae

- ***Non-motile:*** *no flagella or basal bodies at any stage; grow toward food.* ***Nuclear mitosis:*** *nuclear envelope does not break; spindle forms inside nucleus.*
- ***Reproduction:*** *asexual by budding (yeast), fragmentation, mitotic spores; sexual by fusion of specialized cells (genetic diversity).*

Why fungi form a separate kingdom

Like plants	*Like animals*	*Differ from plants*
Cell wall; no centriole; non-motile; spores and sporangia	*Heterotrophs; chitin walls (not cellulose)*	*No chlorophyll; no flagella at any stage; saprotrophs*

- *Hence fungi are neither plants nor animals. More than 100,000 known species; ancestry about 570 million years old; thought to have evolved from red algae (both lack flagella).*

7.4 Classification of fungi (SLO 4)

- *Four main groups by reproduction and sexual structures (and hyphae type).*

Phylum	*Diagnostic features*
Zygomycota (conjugating fungi)	*Fast growing; saprotrophs on bread, vegetables, fruits; non-septate branching hyphae. Asexual: sporangia with spores (Rhizopus nigricans, black bread mould). Sexual: conjugation between + and - strains; zygospore (thick walled).*

Follow Studybeesofficialpak on YouTube Studybeesofficial

Phylum	Diagnostic features
Ascomycota (sac fungi)	Largest group: yeasts, morels, truffles, many plant pathogens. Septate hyphae with perforated septa. Sexual spores (ascospores) in sac-like **asci** inside **ascocarp**. Asexual: **conidia** on conidiophores (no sporangia); yeast by budding. Yeast, Neurospora.
Basidiomycota (club fungi)	Mushrooms, rusts, smuts, puffballs, toadstools, shelf fungi. Septate hyphae. Sexual: **basidia** (club shaped) with 4 **basidiospores**; basidiocarp = fruiting body. Agaricus.
Deuteromycota (imperfect fungi)	No sexual stage known; asexual conidiospores; ancestry from sac fungi; saprophytic or parasitic; parasexuality gives genetic recombination. Penicillium, Aspergillus, Alternaria, Fusarium.

Rhizopus (Zygomycota) life cycle

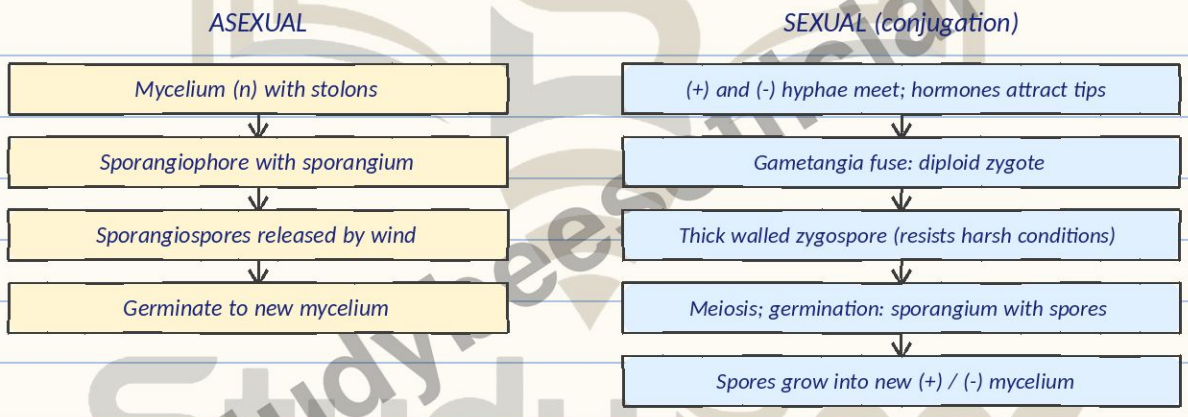


Fig: Reproduction in Rhizopus (Zygomycota)

Ascomycota sexual reproduction

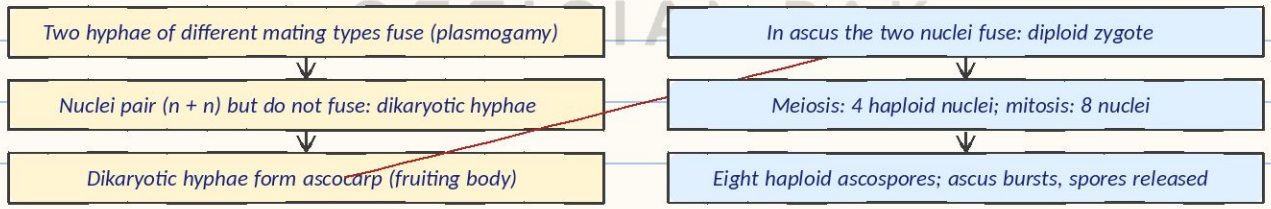


Fig: Sexual reproduction in Ascomycota (sac fungi)

- Yeast: asexual by budding (bud separates) or fission. Cup fungi: cup shaped

Follow Studybeesofficialpak on YouTube Studybeesofficial

ascocarp; morels: stalked; molds: flask shaped.

Basidiomycota (mushroom)



Fig: Mushroom (Agaricus) structure and life cycle

- *Primary mycelium is monokaryotic (n); fusion of different mating types gives secondary dikaryotic (n+n) mycelium (nuclei pair, do not fuse). Compact buttons grow into basidiocarp (stalk + cap with gills). On gills nuclei fuse (zygote), meiosis gives four basidiospores.*

7.5 Importance of fungi (SLO 5, 6, 9)

- *Decompose organic matter; mycorrhizal symbiosis; sources of antibiotics, enzymes, food; research and biotechnology; indicators of environmental health; cause diseases of humans, animals and crops.*

Yeast - baking, brewing, research (SLO 5)

- ***Saccharomyces cerevisiae:** unicellular fungus that ferments sugar to **ethyl alcohol and CO_2**.*
- ***Brewing:** converts sugars of malted grains into alcohol and CO_2: beer and wine flavours and alcohol.*
- ***Baking:** leavening agent; CO_2 trapped in dough makes bread rise (fluffy texture); also pastries and cakes.*
- ***Genetic research:** model organism (simple well-characterized genome, similar to human genes): mutations, recombination, segregation. **Biotechnology:** enzymes, biofuels, pharmaceuticals; yeast takes up foreign DNA, has plasmids*

Follow Studybeesofficialpak on YouTube Studybeesofficial

as vectors; hepatitis B vaccine and interferons; food: cheese, yogurt, soy sauce.

Antibiotics from fungi (SLO 6)

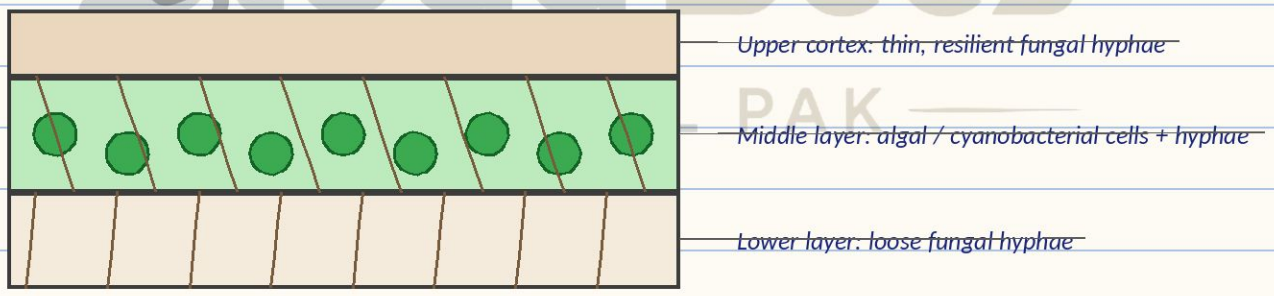
Name of fungus	Antibiotic
Penicillium chrysogenum	Penicillin
Streptomyces griseus	Streptomycin
Streptomyces aureofaciens	Tetracycline
Saccharopolyspora erythraea	Erythromycin
Aspergillus fumigatus	Fumigatin

Mutualism: lichens and mycorrhizae (SLO 7)

Mutualism: *Association in which both partners benefit.*

Lichens

- *Association of a **fungus** (mostly Ascomycota and imperfect fungi, a few Basidiomycota) with a **cyanobacterium or green alga**. Layers: thin upper layer of resilient hyphae; middle layer of hyphae intertwined with photosynthetic cells; bottom layer of loose hyphae.*
- *Traditional view: fungus gets food from algae; algae protected from drying. Recent research: a controlled form of parasitism by the fungus on the algal cells.*



Lichen = fungus (mostly Ascomycota) + alga / cyanobacterium

Fig: Cross section of a lichen

Mycorrhizae

- *Mutually beneficial partnership between **soil fungi and roots**, in 95% of plant families. Hyphae increase soil contact and surface area and take up*

Follow Studybeesofficialpak on YouTube Studybeesofficial

***phosphorus, zinc, copper**; the plant supplies carbohydrates. Plants with mycorrhizae grow better.*

- ***Endomycorrhizae:** penetrate outer root cells, forming coils, swellings and tiny branches, and extend into soil. **Ectomycorrhizae:** form a protective layer outside the root, growing between cell walls; with trees like pines and firs.*

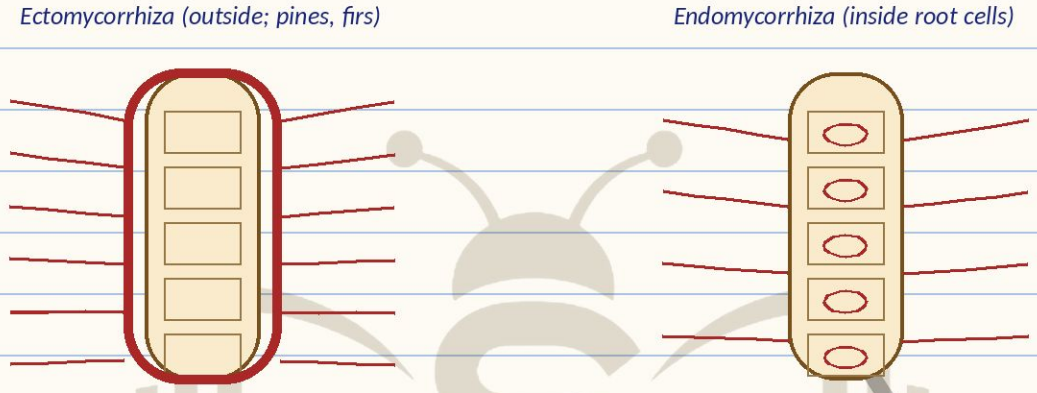


Fig: Types of mycorrhizae

Edible fungi (SLO 8)

- ***Aspergillus tamarii** and other imperfect fungi ferment soybeans into **soy sauce** (Orient). About 200 kinds of edible Basidiomycota mushrooms (mainly **Agaricus**, some cultivated) and about 70 poisonous toadstools. **Morchella esculenta** (morel) and **truffles** (underground fruiting bodies) are Ascomycota.*
- *Edible and poisonous mushrooms look alike; only experts can tell. Deadliest: **Amanita virosa** (destroying angel) and **Amanita phalloides** (death cap); one mushroom can be fatal; some cause hallucinations.*

Ecological impact of fungi (SLO 9)

- ***Decomposers** and nutrient cycling (leaves, wood; enrich soil). **Mycorrhizal partnerships** boost nutrient uptake. **Predatory fungi** regulate microscopic organisms and compete with bacteria in soil. **Biotransformation and pollutant degradation. Carbon cycling and humus** formation (CO_2 released during decomposition).*

Follow Studybeesofficialpak on YouTube Studybeesofficial

Pathogenic role of fungi (SLO 10)

Plant diseases

- *Cause leaf spots, wilting, rotting; epidemics can destroy whole crops.* ***Ascomycota:*** *powdery mildew, chestnut blight, Dutch elm disease, apple scab, brown rot of cherries, peaches, plums, apricots (Monilinia fruticola), ergot of rye (Claviceps purpurea).* ***Basidiomycota:*** *smuts and rusts of wheat and corn.*

Animal and human diseases

Type	Examples
Superficial (skin, hair, nails)	Ringworm (Microsporum audouinii); athlete's foot (Tinea pedis).
Candidiasis	Candida yeast: oral thrush, skin and ear infections.
Systemic	Histoplasmosis (inhaled Histoplasma spores, serious lung infection); aspergillosis (Aspergillus fumigatus; parrots and immunocompromised humans).
Aflatoxin	Aspergillus flavus in badly stored peanuts, corn: cancer-causing mycotoxin.
Ergotism	Ergot of rye: nervous spasm, convulsions, delusions, gangrene.
Pneumocystis pneumonia	Pneumocystis jirovecii in people with weak immunity, e.g. AIDS.

STEAM activity 7.1: Rhizopus on stale bread

- *Moisten stale bread with a spray, keep it in a container or sealed bag for a few days, observe mycelium, sporangia and spores with a magnifying glass or microscope, record drawings and discuss the Rhizopus life cycle and ecological role in decomposition.*

Follow Studybeesofficialpak on YouTube Studybeesofficial

Biology Class 11 National Book Foundation FBISE Quality Notes
Studybeesofficial

Chapter 7: Short Questions with Answers (SLO based)

Q1. [SLO 1] What are protists? Why is the name given?

Ans: Diverse eukaryotic, mostly aquatic, unicellular or simple multicellular organisms. Greek "protist" means the very first, as they were among the first eukaryotes.

Q2. [SLO 1] Why are protists said to be defined by exclusion?

Ans: Any eukaryote that is not a plant, animal or fungus is placed in Protista, so the group is defined by what it is not.

Q3. [SLO 1] What is meant by polyphyletic origin?

Ans: A group whose members do not share a common ancestor but evolved independently from different ancestral sources, as protists did.

Q4. [SLO 1] How do protists differ from bacteria?

Ans: Protists have eukaryotic cells (nucleus, mitochondria, plastids); bacteria are prokaryotes.

Q5. [SLO 1] Give three modes of nutrition in protists.

Ans: Photosynthesis (algae), absorption (water moulds), ingestion of food (protozoa, slime moulds).

Q6. [SLO 1] How do protists move?

Ans: By amoeboid motion (pseudopodia), cilia or flagella, or a combination.

Q7. [SLO 2] Name the four major groups of protists.

Ans: Protozoa, algae, Myxomycota and Oomycota.

Q8. [SLO 2] What are protozoa? Give examples.

Ans: Animal-like unicellular protists (proto = first, zoa = animals): Amoeba, Paramecium, Plasmodium, Entamoeba.

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q9. [SLO 2] Write the functions of vacuoles in protozoa.

Ans: Food vacuoles digest food; contractile vacuoles remove excess water (osmoregulation); storage vacuoles store waste or nutrients.

Q10. [SLO 2] How do protozoa survive unfavourable conditions?

Ans: By forming resistant cysts.

Q11. [SLO 2] Name the locomotory organs of Amoeba, Paramecium and Trypanosoma.

Ans: Pseudopodia, cilia and flagella respectively.

Q12. [SLO 2] What are zooflagellates? Give an example.

Ans: Protozoa that move with flagella; Giardia intestinalis causing giardiasis.

Q13. [SLO 2] What are choanoflagellates?

Ans: Sessile marine / freshwater flagellates anchored by a stalk, with a flagellum surrounded by a collar like sponge collar cells.

Q14. [SLO 2] Differentiate between foraminifera and actinopods.

Ans: Both marine shelled protozoa; forams have calcium shells, actinopods have silica shells.

Q15. [SLO 2] What are apicomplexans?

Ans: Parasitic protozoa with complex life cycles and structures to invade host cells, e.g. Plasmodium.

Q16. [SLO 2] What is leishmaniasis?

Ans: A skin disease caused by Leishmania, spread by sandfly bite: painful sores, fever, fatigue.

Q17. [SLO 2] Why are algae important?

Ans: They perform about 50-60% of the earth's photosynthesis.

Q18. [SLO 2] What are coenocytes? What is a thallus?

Ans: Coenocytes: filamentous algae with many nuclei in one cell without cross walls. Thallus: branched or leaf-like body of seaweeds, without roots, stems and leaves.

Studybeesofficial Study Bees OFFICIAL PAK

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q19. [SLO 2] How do algae differ from plants?

Ans: Algae have unicellular sex organs, their zygote is not protected by the parent and no embryo is formed; no roots, stems, leaves or vascular tissue.

Q20. [SLO 2] Name the six phyla of algae.

Ans: Euglenoids, dinoflagellates, diatoms, brown algae, red algae, green algae.

Q21. [SLO 2] Why are euglenoids unusual?

Ans: About one third have chloroplasts; the others lack them and ingest or absorb food (Euglena).

Q22. [SLO 2] What are diatoms? Describe the structure.

Ans: Most numerous unicellular algae of the oceans; the silica cell wall has two halves like a box, the larger half acting as a lid.

Q23. [SLO 2] What are kelps?

Ans: Massive brown algae up to 75 m long in cool marine waters.

Q24. [SLO 2] What is a holdfast?

Ans: The base of red algae by which they are anchored to rocks or surfaces.

Q25. [SLO 2] Why are green algae believed to be related to the first plants?

Ans: Both have cellulose cell walls and chlorophyll a and b.

Q26. [SLO 2] What is a plasmodium?

Ans: The diploid multinucleate cytoplasmic mass wrapped in a slime sheath, the feeding stage of plasmodial slime moulds.

Q27. [SLO 2] Define sporangium.

Ans: A reproductive structure (Gk spora = seed, angeion = vessel) that produces spores by meiosis.

Q28. [SLO 2] Differentiate between Oomycota and fungi.

Ans: Oomycota have cellulose cell walls and flagellated zoospores; fungi have chitin walls and no flagella. Both have hyphae.

Q29. [SLO 2] What are zoospores and oospores?

Ans: Zoospores: asexual swimming spores with one flagellum. Oospores: thick walled zygotes from oogonium and antheridium for survival.

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q30. [SLO 2] Name the gametangia of Oomycota.

Ans: Female oogonium (1-8 eggs) and male antheridium (many male nuclei).

Q31. [SLO 3] What is a red tide? What does it cause?

Ans: A harmful algal bloom of toxic dinoflagellates (Alexandrium); shellfish accumulate the neurotoxin and cause paralytic shellfish poisoning in humans.

Q32. [SLO 3] What are frustules? State their use.

Ans: Silica shells of diatoms, used in cosmetics, pharmaceuticals and nanotechnology.

Q33. [SLO 3] What are the uses of brown and red algae?

Ans: Brown: edible kelp, alginates as thickeners, protect coasts. Red: food, cosmetics, agar for culture media.

Q34. [SLO 3] Why is Chlorella important?

Ans: Used in photosynthesis research, a high-protein health food and a potential biofuel source.

Q35. [SLO 3] What disease does Entamoeba histolytica cause?

Ans: Amoebic dysentery: severe diarrhoea, abdominal pain, fever, from contaminated food or water.

Q36. [SLO 3] What is Acanthamoeba keratitis?

Ans: An eye infection caused by Acanthamoeba from improper contact lens hygiene or contaminated water.

Q37. [SLO 3] What did Phytophthora infestans cause?

Ans: The Irish potato famine; late blight of potato, controlled by destroying infected plants before lifting tubers.

Q38. [SLO 4] What is mycology? Define mycelium and hyphae.

Ans: Study of fungi. Hyphae are filaments; the network of hyphae is the mycelium (thallus).

Q39. [SLO 4] Differentiate between septate and aseptate hyphae.

Ans: Septate hyphae have cross walls (Penicillium); aseptate (coenocytic) hyphae have none and are multinucleate (Rhizopus).

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q40. [SLO 4] What is the chemical nature of the fungal cell wall and reserve food?

Ans: Cell wall of chitin; reserve food is glycogen, not starch.

Q41. [SLO 4] How do fungi obtain food?

Ans: They are heterotrophs with extracellular digestion: enzymes break down organic matter and the products are absorbed.

Q42. [SLO 4] Why are fungi placed in a separate kingdom?

Ans: They have chitin walls, no chlorophyll, no flagella and are absorptive heterotrophs: neither plants nor animals.

Q43. [SLO 4] Which is the largest living organism?

Ans: The fungus Armillaria ostoyae (honey mushroom), covering over 2385 acres in Oregon.

Q44. [SLO 4] What is nuclear mitosis in fungi?

Ans: The nuclear envelope does not break down; the spindle forms inside the nucleus and the envelope constricts between the daughter chromosome clusters.

Q45. [SLO 4] Name the four phyla of fungi.

Ans: Zygomycota, Ascomycota, Basidiomycota and Deuteromycota.

Q46. [SLO 4] Why are Zygomycota called conjugating fungi?

Ans: Their sexual reproduction is by conjugation between + and - strains forming a zygospore.

Q47. [SLO 4] What is a zygospore?

Ans: A thick walled resistant zygote of Zygomycota that survives unfavourable conditions.

Q48. [SLO 4] What are asci and ascocarp?

Ans: Asci are sac-like cells containing eight ascospores; ascocarp is the fruiting body that holds them.

Q49. [SLO 4] What are conidia?

Ans: Asexual spores of Ascomycota formed on tips of conidiophores and carried by wind.

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q50. [SLO 4] What are basidia and basidiocarp?

Ans: Basidia: club shaped cells bearing four basidiospores. Basidiocarp: the mushroom fruiting body (stalk + cap with gills).

Q51. [SLO 4] What is dikaryotic (n+n) mycelium?

Ans: Hyphae whose cells contain two nuclei from different parents that pair but do not fuse.

Q52. [SLO 4] Why are Deuteromycota called imperfect fungi?

Ans: No sexual stage is known; they reproduce only asexually by conidiospores. Examples: Penicillium, Aspergillus.

Q53. [SLO 4] What is parasexuality?

Ans: Genetic recombination in imperfect fungi by exchange of chromosome portions between two nuclei in the same hypha.

Q54. [SLO 5] Why is yeast important in baking and brewing?

Ans: It ferments sugar to ethyl alcohol and CO_2: CO_2 makes dough rise; alcohol is used in beer and wine.

Q55. [SLO 5] Why is yeast a model organism in genetic research?

Ans: It has a simple well characterized genome with similarity to human genes; studies mutations, recombination and segregation.

Q56. [SLO 5] Name two proteins made with yeast.

Ans: Hepatitis B vaccine (yeast alone) and interferons.

Q57. [SLO 6] Name fungi from which antibiotics are obtained.

Ans: Penicillium chrysogenum (penicillin), Streptomyces griseus (streptomycin), S. aureofaciens (tetracycline), Saccharopolyspora erythraea (erythromycin), Aspergillus fumigatus (fumigatin).

Q58. [SLO 7] What is mutualism?

Ans: An association in which both partners get benefit, e.g. lichens and mycorrhizae.

Q59. [SLO 7] What are lichens?

Ans: Associations of a fungus (mostly Ascomycota) with a cyanobacterium or green alga.

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q60. [SLO 7] Describe the layers of a lichen.

Ans: Thin upper layer of resilient hyphae, middle layer of hyphae with photosynthetic cells, lower layer of loose hyphae.

Q61. [SLO 7] What is a mycorrhiza? Name its types.

Ans: Mutually beneficial partnership between soil fungi and plant roots; endomycorrhizae and ectomycorrhizae.

Q62. [SLO 7] How do mycorrhizae benefit plants?

Ans: Hyphae increase surface area and absorb phosphorus, zinc and copper; plants grow better. The fungus gets carbohydrates.

Q63. [SLO 8] Name edible fungi.

Ans: Agaricus mushrooms, Morchella esculenta (morel), truffles, and Aspergillus tamarii used for soy sauce.

Q64. [SLO 8] Name two deadly mushrooms.

Ans: Amanita virosa (destroying angel) and Amanita phalloides (death cap).

Q65. [SLO 9] Write the ecological role of fungi.

Ans: Decomposers and nutrient recyclers, mycorrhizal partners, predators of microbes in soil, pollutant degraders and humus formers.

Q66. [SLO 10] Name three fungal diseases of plants.

Ans: Powdery mildew, Dutch elm disease, smuts and rusts (also apple scab, ergot of rye).

Q67. [SLO 10] What are ringworm and athlete's foot?

Ans: Superficial fungal skin infections: ringworm by Microsporum audouinii, athlete's foot by Tinea pedis.

Q68. [SLO 10] What is aflatoxin?

Ans: A cancer-causing mycotoxin from Aspergillus flavus in improperly stored peanuts and corn.

Q69. [SLO 10] What is ergotism?

Ans: Disease from ergot of rye (Claviceps purpurea) causing spasms, convulsions, delusions and gangrene.

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q70. [SLO 10] What are candidiasis and histoplasmosis?

Ans: Candidiasis: Candida yeast infection like oral thrush. Histoplasmosis: serious lung infection from inhaling Histoplasma spores.

Studybeesofficial
Study Bees
OFFICIAL PAK

Follow Studybeesofficialpak on YouTube Studybeesofficial

Biology Class 11 National Book Foundation FBISE Quality Notes
Studybeesofficial

Chapter 7: Long Questions with Answers (SLO based)

Q1. [SLO 1] Explain protists as a diverse group of eukaryotes with polyphyletic origin, defined by exclusion.

Ans:

- *Protists: up to 200,000 species; eukaryotic; unicellular, colonial, coenocytic or simple multicellular without tissues.*
- *Defined by exclusion: any eukaryote that is not plant, animal or fungus. Polyphyletic: many different ancestors (maybe about twenty lineages).*

Feature	*Diversity*
Nutrition	*Algae autotrophs; water moulds absorptive; protozoa, slime moulds ingestive*
Habitat	*Water, soil, symbionts, parasites, extremophiles*
Locomotion	*Pseudopodia, cilia, flagella*
Reproduction	*Asexual or sexual (meiosis and syngamy); no embryos*

Q2. [SLO 2] Describe the salient features of protozoa with examples.

Ans:

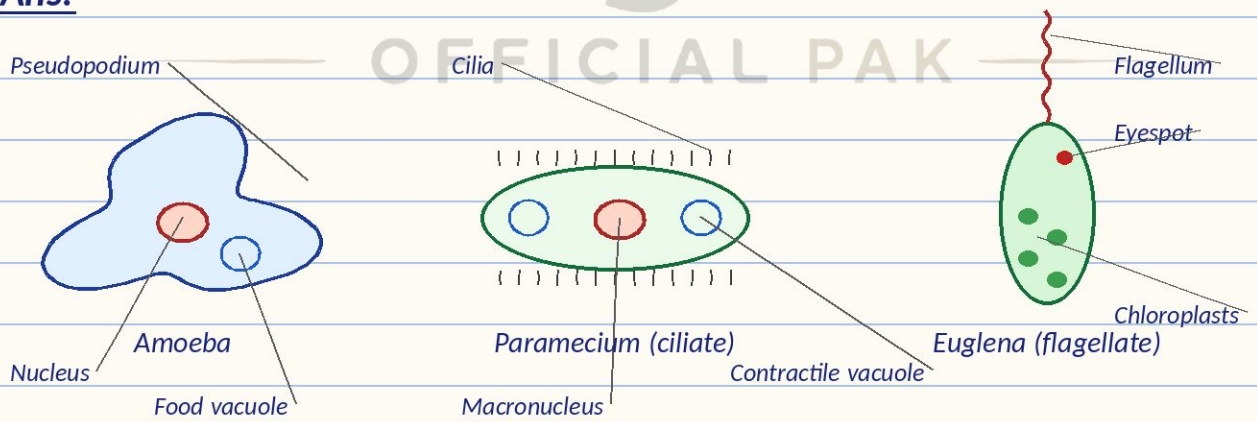


Fig: Amoeba, Paramecium and Euglena

- *Unicellular, animal-like; aquatic (Amoeba, Paramecium, Radiolaria), parasitic*

Follow Studybeesofficialpak on YouTube Studybeesofficial

(Plasmodium, Entamoeba) or in soil (Vorticella). Food, contractile and storage vacuoles; locomotion by pseudopodia, cilia or flagella; asexual and sexual reproduction; regeneration; cysts.

- *Groups: amoebas, zooflagellates (Giardia), choanoflagellates, ciliates (Paramecium, Stentor), foraminifera / actinopods (shells), apicomplexans (Plasmodium).*

Q3. [SLO 2] Describe the salient features of algae and the six phyla with examples.

Ans:

- *Photosynthetic (50-60% of earth's photosynthesis); unicellular, filamentous (coenocytic) or thallus; no roots, stems, leaves; pigments chlorophyll a, carotenoids, xanthophyll, phycoerythrin; flagellated stages except red algae; unicellular sex organs.*

Phylum	Features
Euglenoids	Freshwater; 1/3 with chloroplasts; Euglena
Dinoflagellates	Cellulose plates, two flagella, red tides
Diatoms	Silica box-like wall (frustules)
Brown algae	Kelps up to 75 m, cool seas
Red algae	Warm seas, holdfast, agar
Green algae	Cellulose, chl a and b; Spirogyra, Volvox, Chlorella

Q4. [SLO 2] Describe the characteristics and life cycle of Myxomycota (slime moulds).

Ans:

- *Plasmodial slime moulds exist as plasmodium: diploid, multinucleate, with slime sheath; fungus-like protists with changing forms.*

Follow Studybeesofficialpak on YouTube Studybeesofficial

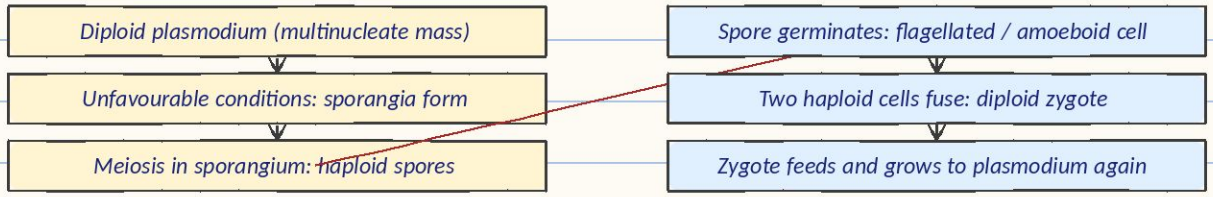


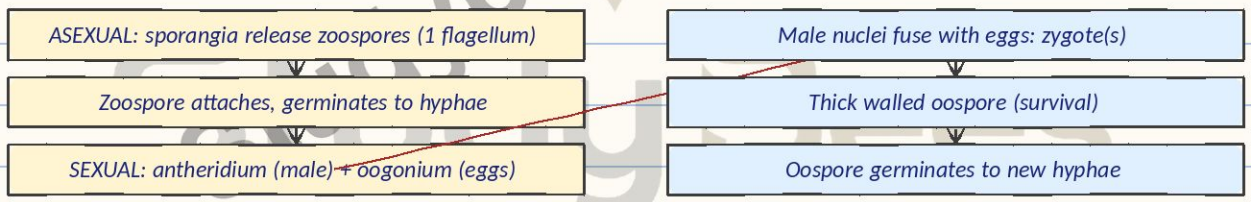
Fig: Life cycle of plasmodial slime mould (Physarum)

- *Unfavourable conditions: sporangia; meiosis gives haploid spores; spores release flagellated / amoeboid haploid cells which fuse to a diploid zygote that grows into plasmodium (Physarum).*

Q5. [SLO 2] Describe the characteristics and life cycle of Oomycota (water moulds).

Ans:

- *Water moulds, white rusts, downy mildews; aquatic or moist habitats; parasites or saprotrophs; cellulose wall; aseptate coenocytic hyphae.*



Studybeesofficial OFFICIAL PAK

Fig: Life cycle of Oomycota (Phytophthora infestans)

- *Asexual: zoospores from sporangia. Sexual: oogonium + antheridium, oospore. Examples: Phytophthora infestans, Saprolegnia.*

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q6. [SLO 3] Justify how protists are important for humans.

Ans:

- *Useful: diatoms (oxygen, food webs, frustules in industry), brown algae (food, alginates, coast protection), red algae (agar, food, cosmetics), green algae (Chlorella food, biofuel), dinoflagellates (producers).*
- *Harmful: Plasmodium (malaria), Entamoeba histolytica (amoebic dysentery), Acanthamoeba (keratitis), Leishmania, Phytophthora (potato blight, Irish famine), toxic dinoflagellates (paralytic shellfish poisoning).*

Q7. [SLO 4] Describe the general characteristics of fungi and explain why they are placed in a separate kingdom.

Ans:

- *Eukaryotic, heterotrophic by extracellular digestion; chitin cell wall; glycogen reserve; mycelium of hyphae (septate / aseptate); non-motile (no flagella); reproduce asexually (budding, fragmentation, spores) and sexually.*

Like plants	***Like animals***
Cell wall, non-motile, no centriole, spores and sporangia	*Heterotrophs; chitin not cellulose*

- *Differ from plants: no chlorophyll, no flagella, saprotrophic. So fungi form a separate kingdom.*

Q8. [SLO 4] Classify fungi into phyla and give the diagnostic features of each.

Ans:

Phylum	***Features***
Zygomycota	*Non-septate hyphae; sporangia (asexual); conjugation and zygospore; Rhizopus*
Ascomycota	*Septate; asci with ascospores in ascocarp; conidia; yeast, Neurospora, morels*

Follow Studybeesofficialpak on YouTube Studybeesofficial

Phylum	Features
Basidiomycota	Septate; basidia with basidiospores on basidiocarp; mushrooms, rusts, smuts
Deuteromycota	Imperfect: no sexual stage; conidia; Penicillium, Aspergillus

Q9. [SLO 4] Describe the life cycle of Rhizopus (Zygomycota).

Ans:

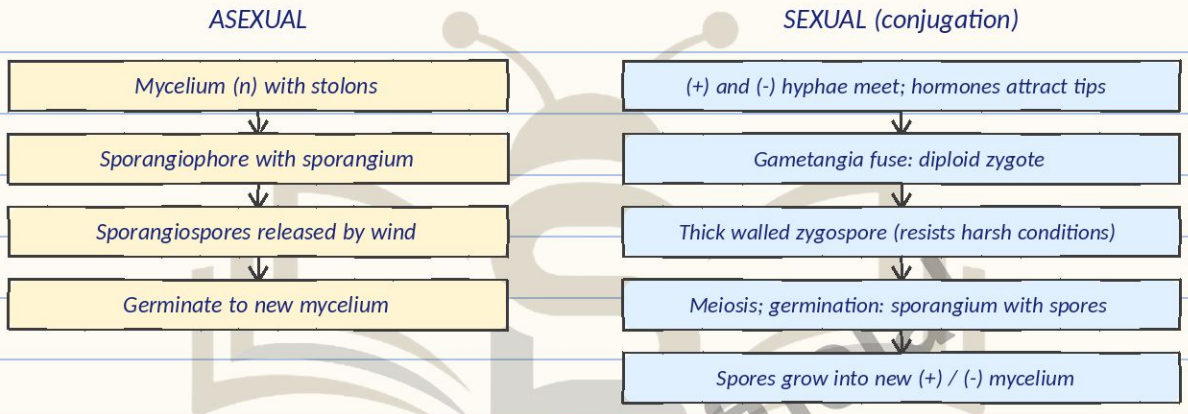


Fig: Reproduction in Rhizopus (Zygomycota)

- Asexual: sporangia with spores. Sexual: + and - hyphae, hormones, gametangia, diploid zygote, thick zygospore, meiosis on germination, sporangium with spores.

Q10. [SLO 4] Describe reproduction in Ascomycota and Basidiomycota.

Ans:

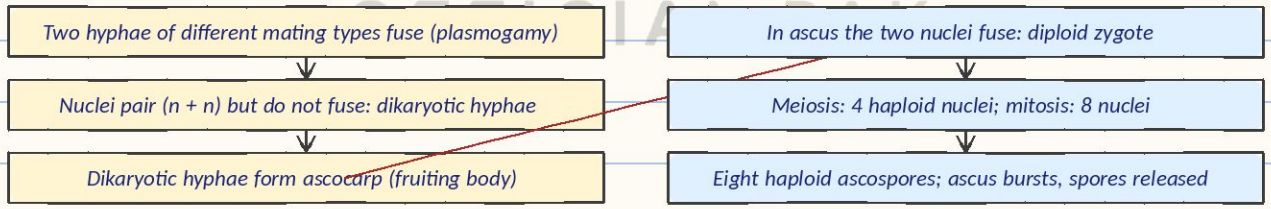


Fig: Sexual reproduction in Ascomycota (sac fungi)

- Ascomycota: asexual by conidia (yeast by budding / fission); sexual: dikaryotic

Follow Studybeesofficialpak on YouTube Studybeesofficial

hyphae, ascocarp, ascus: fusion, meiosis, mitosis, eight ascospores.

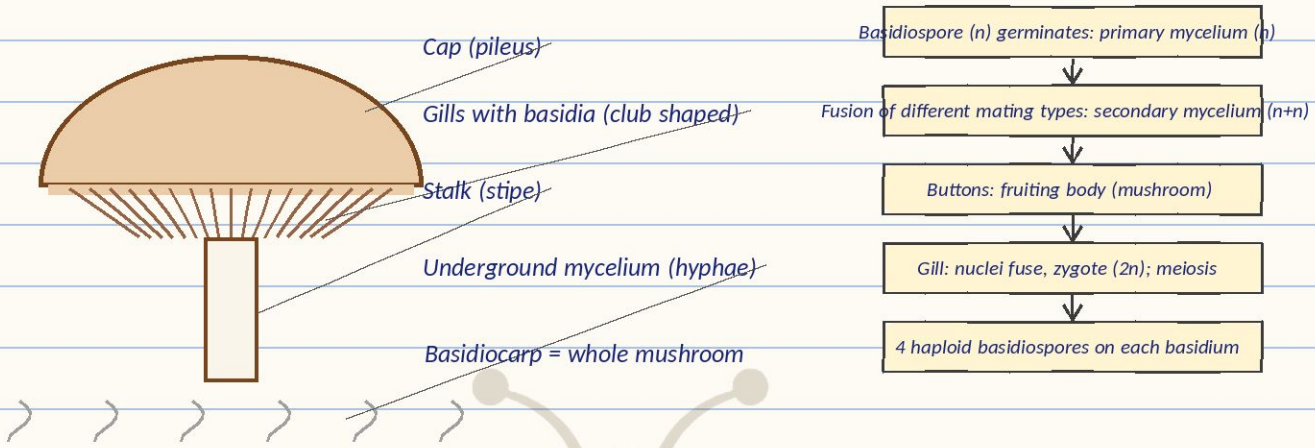


Fig: Mushroom (Agaricus) structure and life cycle

- *Basidiomycota: primary mycelium (n), secondary dikaryotic mycelium (n+n), buttons, basidiocarp (mushroom), basidia: fusion, meiosis, four basidiospores.*

Q11. [SLO 5] Explain yeast as a unicellular fungus used in baking and brewing and in genetic research.

Ans:

- *Saccharomyces cerevisiae ferments sugar to ethyl alcohol and CO_2.*
- *Baking: CO_2 leavens dough. Brewing: beer and wine. Research: model organism, simple genome similar to human genes; recombination, mutation. Biotechnology: enzymes, biofuels, hepatitis B vaccine, interferons; plasmids as vectors.*

Studybeesofficial
Study Bees
OFFICIAL PAK

Q12. [SLO 6] Name fungi from which antibiotics are obtained. Write notes on their importance.

Ans:

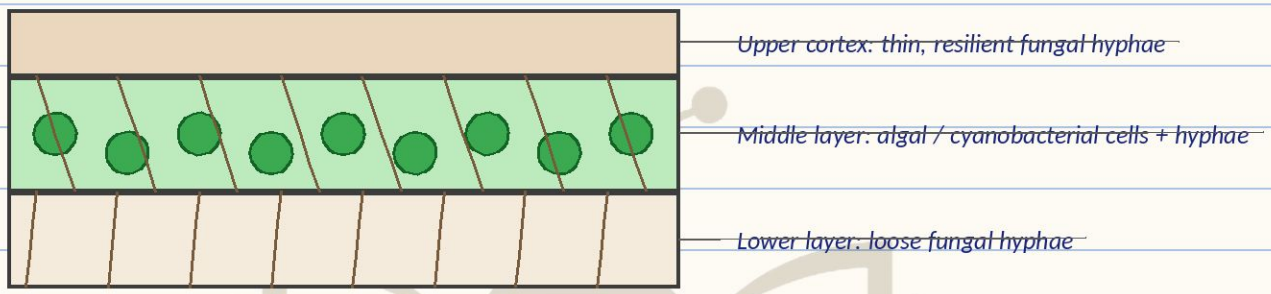
Fungus	***Antibiotic***
Penicillium chrysogenum	*Penicillin*
Streptomyces griseus	*Streptomycin*
Streptomyces aureofaciens	*Tetracycline*
Saccharopolyspora erythraea	*Erythromycin*

Follow Studybeesofficialpak on YouTube Studybeesofficial

Fungus	Antibiotic
Aspergillus fumigatus	*Fumigatin*

Q13. [SLO 7] Explain the mutualism established in lichens and mycorrhizae.

Ans:

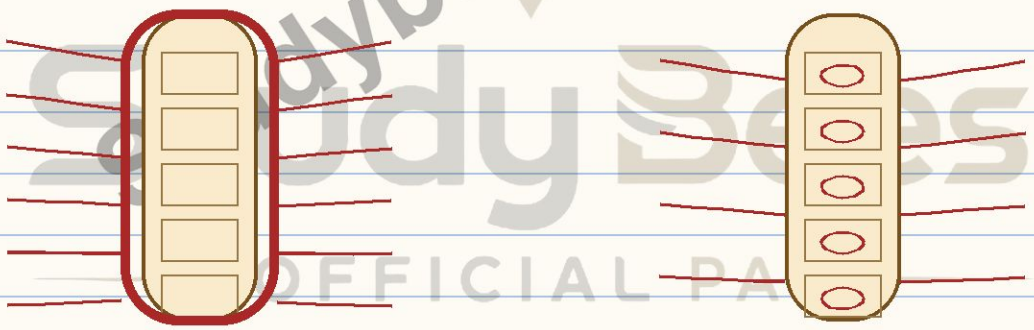


Lichen = fungus (mostly Ascomycota) + alga / cyanobacterium

Fig: Cross section of a lichen

- Lichen: fungus + alga / cyanobacterium; fungus gets food, alga gets protection from drying (recent view: controlled parasitism). Layers: upper hyphae, algal layer, loose hyphae.

Ectomycorrhiza (outside; pines, firs) Endomycorrhiza (inside root cells)



Fungus gets sugars; plant gets phosphorus, zinc, copper and water (95% of plant families)

Fig: Types of mycorrhizae

- Mycorrhiza: soil fungi + roots (95% of plant families): hyphae absorb phosphorus, zinc, copper; plant gives carbohydrates. Ecto- (outside root; pines, firs) and endomycorrhizae (inside outer root cells).

Follow Studybeesofficialpak on YouTube Studybeesofficial

Q14. [SLO 8, 9] Give examples of edible fungi and describe the ecological impact of fungi.

Ans:

- *Edible: Agaricus mushrooms (about 200 edible kinds), morels, truffles (Ascomycota), Aspergillus tamarii for soy sauce. Deadly: Amanita virosa, A. phalloides.*
- *Ecological: decomposition and nutrient cycling; mycorrhizal partnerships; predation in soil; pollutant degradation; carbon cycling and humus formation.*

Q15. [SLO 10] Discuss the pathogenic role of fungi in plants and animals.

Ans:

- *Plants: powdery mildew, chestnut blight, Dutch elm disease, apple scab, brown rot, ergot of rye (Ascomycota); smuts and rusts of wheat and corn (Basidiomycota).*
- *Animals / humans: ringworm, athlete's foot, candidiasis (oral thrush), histoplasmosis, aspergillosis, aflatoxin (A. flavus), ergotism, Pneumocystis pneumonia in AIDS patients.*

Studybeesofficial Study Bees OFFICIAL PAK

Follow Studybeesofficialpak on YouTube Studybeesofficial